Track 1 technical documentation

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MVA 2026 - Track 1 Variant-Prioritization Pipeline - Technical Documentation

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1. Purpose & scope

The Track 1 pipeline consumes a whole-genome sequencing (WGS) variant call format (VCF) file from a single proband, applies a deterministic series of annotation, rarity, phenotype-similarity and scoring steps, and produces a ranked table of candidate variant pairs together with a **challenge-compliant submission CSV**.

The end-to-end result reported in *REPORT.md* is:

- **6,962** ranked candidate rows, the clinically confirmed compound-heterozygous genotype of **BUB1B** ranked **1** with an **EPCR** score of **0.5823**.
- The top-ranked pair consists of:
  - *BUB1B p.Leu737Ter* (chr15:40,209,701 T>G, stop-gained, ClinVar allele ID 529272)
  - *BUB1B p.Asn1002Lys* (chr15:40,220,612 T>G, missense)
- The second-ranked EPCR score is **0.5243**; a weight of **0 -> 2.0** on the HPO term changes the BUB1B pair's rank from **14** to **1** (see Section 4; the running table and headline scores are identical to the September-corrected head).

No external APIs are invoked; all reference data are local files configured via `pipeline/conf/track1_config.yml`.

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2. Execution environment & dependencies

Component	Used by (`pipeline/scripts/` under root)	Role	Docs
Python 3.12	-	Interpreter	docs.python.org/3
cyvcf2	`prioritize.py::iter_qc_variants` (VCF streaming)	fast VCF parsing, genotype fields	brentp.github.io/cyvcf2
pysam	`vcf.py::GnomadAfJoin`, `::ClinVarJoin`; `annotate.py::LocalAnnotator`	HTSlib wrapper; FASTA, Tabix	pysam.readthedocs.io
PyYAML	`prioritize.py::load_conf`; self-tests (`annotate.py`, `_hpo.py`)	YAML configuration parsing	pyyaml.org/docs
matplotlib (Agg)	`docs/figures/pipeline_figure.py`	overview-figure graphics	matplotlib.org
pandoc 2.19 or later	`docs/build_pdf.sh`	Markdown → PDF conversion	pandoc.org/MANUAL

No other third-party libraries (e.g. pandas, NumPy) are imported by the pipeline code.

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3. Data flow - stage-by-stage description

The pipeline execution obeys the flow shown in **Figure 1** (generated by `pipeline_figure.py`). Each stage lists the input artifact, the script and principal functions, the library calls used, and the output artifact.

3.1 Input VCF

- **Artifact:** `pipeline/conf/track1_config.yml` -> `data.vcf` points to `~/hf-rare/data/WGS_EX2312012_HGWCNDSX7.vcf.gz` (5,012,204 records, GRCh38, bare contig names).
- **No code** is executed; the file is read by **cyvcf2.VCF** in the next stage.

3.2 Quality-control filtering

- **Script:** `pipeline/scripts/prioritize.py`
- **Function:** `iter_qc_variants(vcf_path, qc)`
- **Libraries:** `cyvcf2.VCF` (iteration), Python built-ins for numeric checks.
- **Logic:**
  - Keep records where `FILTER` is `PASS` or `None`.
  - Require genotype quality **GQ >= 20**, read depth **DP >= 10**, alternate-allele fraction **>= 0.25** of the **called allele** (`AD[k]` for each GT-gradient allele; multiallelic records evaluate per allele).
  - Accept records with at least one called alternate allele after those gates (`0/1`, `0/2`, `1/2`, `2/2`, phased variants; haploid = hom-alt; partial or fully missing calls skipped; malformed/missing AD/DP rejected).
- **Counters (reported on STDERR):** `vcf_in=5,012,204`, `qc_pass=4,592,493` (records with >=1 emitted allele; the September gate kept 4,592,140, losing multiallelic records whose ALT1 failed while ALT2 passed; see *REPORT.md*).
- **Output:** filtered `cyvcf2.VCF` record generator passed to the annotation stage.

3.3 Consequence annotation

- **Script:** `pipeline/scripts/annotate.py`
- **Class:** `LocalAnnotator`
- **Key methods:** `__init__`, `_load_gtf`, `_candidates`, `_classify`, `_coding`, `_cadd`, `annotate`
- **Libraries:** `pysam.FastaFile` for reference FASTA, `pysam.TabixFile` for CADD tiles, `bisect` for interval search, standard library (`gzip`, `re`).
- **Reference inputs (see Section 5):**
  - **GRCh38 FASTA** (`refs.genome_fasta`)
  - **GENCODE v47 MANE GTF** (`refs.mane_gtf`)
  - **CADD v1.7 per-chromosome bgz files** (`refs.cadd_dir`).
- **Logic:**
  1. Load MANE GTF into per-chromosome sorted transcript lists (`self.tx`).
  2. For each ALT allele, locate candidate transcripts within `MAX_TX_SPAN = 2,500,000 bp`.

3. Classify splice distance, coding region, and compute codon effect using the `CODON` table.
4. Look up CADD PHRED score via Tabix; if no tile covers the site, the field remains `None`.

- **Counter (STDERR `[instrument]` line, *prioritize.py:*):** `coding_or_splice=12,952` after annotation (see *REPORT.md*).
- **Output:** a list of annotation dictionaries per ALT allele, passed downstream.

3.4 Rarity & ClinVar join

- **Script:** `pipeline/scripts/vcf.py`
- **Classes:** `GnomadAfJoin`, `ClinVarJoin`
- **Functions:** `GnomadAfJoin.af`, `ClinVarJoin.lookup`
- **Libraries:** `pysam.TabixFile` for both gnomAD sites (per-chromosome VCF) and ClinVar VCF; `cyvcf2` indirectly via the VCF iterator.
- **Reference inputs:**
  - **gnomAD v4.1 exomes** (`refs.gnomad_v41_exomes_dir`), accessed by `GNOMAD_PATTERN = "gnomad.exomes.v4.1.sites.chr{chrom}.vcf.bgz"`.
  - **ClinVar VCF** (`refs.clinvar_vcf`).
- **Logic:**
  - **gnomAD:** a forward-only sorted merge cursor (`_ChromCursor`) caches allele-frequency maps per position; missing chromosome files return `None` (treated as neutral). Cache is trimmed when > 500,000 entries.
  - **ClinVar:** exact-allele lookup via tabix; returns `(CLNSIG, CLNDN)` or `(None, None)`. Results are *background* evidence only (not used in scoring weights).
- **Counters (STDERR):** counts of `af_absent`, `af_rare`, `af_common`, `af_no_chrom_file`, and `clinvar_hits` (see *REPORT.md* line 34).
- **Output:** `af` (float or `None`) and optional ClinVar fields attached to each variant record.

3.5 HPO phenotype similarity

- **Script:** `pipeline/scripts/_hpo.py`
- **Class:** `HPOScorer`
- **Key method:** `gene_score(gene)`
- **Libraries:** Standard library (`os`, `re`, `functools.lru_cache`).
- **Reference inputs:**
  - **hp.obo** (`refs.hp_obo`) - HPO ontology.
  - **hpo_gene_to_phenotype.tsv** (`refs.hpo_map`).
- **Logic:**
  1. Parse the OBO file to build a direct-parent map (`self.parents`).
  2. Parse the gene-to-phenotype map (`self.gene_terms`).
  3. `closure(term)` returns the term plus all ancestors (transitive `is_a`).
  4. `gene_score` computes the mean over the eight proband HPO terms (`phenotype.hpo_terms`):
     - **1.0** if the gene's term set contains the proband term (or any ancestor).
     - **0.5** if only a direct parent matches.
     - **0.0** otherwise.
- **Self-test:** confirms positive scores for *BUB1B* and *CEP57*, zero for *ZNF717* (see *_hpo.py* line 90).
- **Integration:** cached per-gene scores are queried in `prioritize.py` (`gene_hpo(gene)`).

3.6 Scoring - WorkPrior model

- **Script:** `pipeline/scripts/prior.py`
- **Classes:** `ScoredVariant`, `WorkPrior`
- **Key method:** `WorkPrior.internal(v)`
- **Libraries:** `math` (for constants) and `dataclasses` (for `ScoredVariant`).
- **Configuration:** `pipeline/conf/track1_config.yml` -> `scoring` block supplies weights:
  - `founder_weight = 0.45`
  - `pathogenicity_weight = 0.60`
  - `rare_weight = 0.30`
  - `hpo_weight = 2.0`
- **Component functions:**
  - `_founder(v)`: LoF heterozygote -> 0.65, het -> 0.45; hom-alternate LoF -> 1.0; missense/inframe have intermediate values (see *prior.py* lines 61-70).
  - `_patho(v)`: LoF -> 1.0, missense/inframe -> 0.5, synonymous -> 0.05, others -> 0.1; optionally upgraded to 1.0 if CADD >= 25 (CADD gate, lines 82-84).
  - `_rare(v)`: `af is None` -> 0.5 (neutral); `af == 0` -> 1.0 (ultrarare); `< max_ultra (0.0001)` -> 0.9; `< max_rare (0.01)` -> 0.7; otherwise -> 0.0.

  - `_hpo(v)`: uses the cached HPO gene score (0 - 1).
- **Scoring formula (exact as implemented):**

```
epcr = max(1e-6, min(1.0,
        (w_f*founder + w_p*patho + w_r*rare + w_h*hpo) /
        (w_f + w_p + w_r + w_h)))
```

where **w_f**, **w_p**, **w_r**, **w_h** are the four weights above. - **Compound-het pair score:** defined in `prior.py` line 120:

```
pair_score = max(a, b) * (0.5 + 0.5 * min(a, b))
```

where `a` and `b` are the internal EPCR scores of the two variants.

3.7 Candidate pairing & ranking

- **Script:** `pipeline/scripts/prioritize.py` (the `main()` function).
- **Workflow:**
  1. Iterate QC-filtered records; annotate each ALT allele via `LocalAnnotator`.
  2. Apply rarity/ClinVar join (if `--with-af` is set).
  3. Compute HPO gene score on-the-fly (`gene_hpo`).
  4. Build a `ScoredVariant` instance and call `WorkPrior.internal` to set `v.internal`.
  5. Collect heterozygous LoF variants per gene (`by_gene`) and hom-alternate singles (`singles`).
  6. **Compound-het pairing:** for each gene with >= 2 heterozygous candidates, pair each LoF het (`het_lof`) with any damaging het (`het_dmg`).
  7. **Singles:** all candidate homozygous-alternate variants (any internal score; the removed >0.3 subset counter is documented in the audit-resolution record; the written rows equal the emitted `hom_singles` count).
  8. Sort pairs by descending pair score, then singles by descending internal score. Write a TSV (`annot_full.tsv`) with 16 columns (see Section 6). The final ranked table contains **6,962** rows (see *REPORT.md* line 8).
- **Counters (STDERR):** `coding_or_splice=12,952`, `genes_with_hets=...`, `compound_het_pairs=2,211`, `hom_singles=4,751` (see *REPORT.md*; the phantom `hom_candidates` >0.3 subset counter was removed in the audit emission-contract fix).

3.8 Submission CSV generation

- **Script:** `pipeline/scripts/make_submission_csv.py`
- **Function:** `main()` (argparse-driven)
- **Logic:**
  - Reads the ranked TSV, skips header/comments, clamps EPCR to `(1e-6, 1.0)`, formats with four decimals.
  - Adds the fixed proband identifier `PROBAND01` and prefixes chromosome names with `chr`.
  - Emits the first **10** rows (or the value of `--rows`) in the official 12-column schema: `proband_id,chrom_1,pos_1,ref_1,alt_1,chrom_2,pos_2,ref_2,alt_2,epcr,finding_type,notes`.
  - The shipped `submission_track1.csv` (see *submission_track1.csv* header) matches this format; the `notes` column is hand-crafted in the reference CSV, while the generator would produce a compact `gene=...;consequence` string.

---

4. Scoring / prioritisation model details

The **WorkPrior** model combines four evidence components into the EPCR score (a bounded heuristic rank score in (0, 1], written into the challenge's EPCR column; it is **not an empirically calibrated probability**, and no calibration study exists supporting reading it as absolute risk):

Component	Weight (config)	Description	Implementation reference
Founder	`founder_weight = 0.45`	Prior probability that a loss-of-function allele exists in the gene (based on zygosity & LoF classification)	`prior.py::_founder`
Pathogenicity	`pathogenicity_weight = 0.60`	Variant effect class (LoF -> 1.0, missense/inframe -> 0.5, synonymous -> 0.05, other -> 0.1) possibly upgraded by CADD	`prior.py::_patho`

Component	Weight (config)	Description	Implementation reference
Rarity	`rare_weight = 0.30`	gnomAD allele frequency tier (ultrarare -> 1.0, rare -> 0.7, common -> 0.0, missing -> 0.5)	`prior.py::_rare`
Phenotype (HPO)	`hpo_weight = 2.0`	Gene-to-phenotype similarity (mean over eight proband HPO terms)	`prior.py::_hpo`

The weighted sum is normalised by the total weight and clamped to `(1e-6, 1.0]` (worked out by WorkPrior.internal in `prior.py`).

Compound-het pair scoring (`compound_het_score`) multiplies the stronger allele's internal score by the mean of the two scores (i.e. `0.5 + 0.5*min`) - a simple, monotonic function that favours pairs where both alleles are high-scoring.

Changing the HPO weight from **2.0** to **0** lowers the BUB1B pair's rank to **14** (EPCR 0.6913) as documented in *REPORT.md* (record `desensitization_audit19b.json`, measured 2026-10-19).

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5. Configuration reference (`track1_config.yml`)

Section	Key	Meaning
`data`	`vcf`	Path to the proband VCF (gzipped).
	`vcf_chr_prefix`	Whether contig names in the VCF include the `chr` prefix (boolean).
	`sample`	Sample identifier (for downstream naming).
	`build`	Reference genome build.
`refs`	`cadd_dir`	Ordered list of directories to search for CADD per-chromosome `.vcf.bgz` files.
	`mane_gtf`	Path to MANE GTF (tabix-indexed).
	`genome_fasta`	Path to GRCh38 FASTA (FAI-indexed).
	`hp_obo`	HPO ontology OBO file.
	`hpo_map`	Gene-to-HPO mapping TSV.
	`gnomad_v41_exomes_dir`	Directory containing per-chromosome gnomAD VCFs.
	`clinvar_vcf`	ClinVar VCF (tabix-indexed).
`variants`	`qc.min_gq`	Minimum genotype quality for QC.
	`qc.min_dp`	Minimum read depth for QC.
	`qc.min_alt_frac`	Minimum alternate-allele fraction for QC.
	`af.max_rare`	Upper AF threshold defining a *rare* allele.
	`af.max_ultrarare`	Upper AF threshold defining an *ultrarare* allele.
	`effect.lof_terms`	Set of consequence strings classified as loss-of-function.
	`effect.missense_terms`	Missense consequence strings.
	`effect.inframe_terms`	In-frame indel consequence strings.
	`effect.exploit_cadd_if_available`	Toggle CADD-based boost (used in `_patho`).
`pairing`	`compound_het_rule`	Human-readable description (not parsed).
	`same_gene_only`	Enforce same-gene pairing (hard-coded in `prioritize.py`).
	`allow_unknown_phase`	Permit unknown phasing (hard-coded).
`phenotype`	`hpo_terms`	List of eight proband HPO term IDs.
`scoring`	`founder_weight`	Weight of the founder component.
	`pathogenicity_weight`	Weight of the pathogenicity component.

Section	Key	Meaning
	`rare_weight`	Weight of the rarity component.
	`hpo_weight`	Weight of the HPO phenotype component.
`output`	`write_dir`	Directory where `annot_full.tsv` is written (default).
	`track1_csv_columns`	List of column names for the final CSV (hard-coded in `make_submission_csv.py`).

Examples from `pipeline/conf/track1_config.yml` (… marks an elided list tail):

```yaml
data.vcf: ~/hf-rare/data/WGS_EX2312012_HGWCNDSX7.vcf.gz
  vcf_chr_prefix: false
  sample: "WGS_EX2312012"
  build: "GRCh38"
refs.cadd_dir: ["~/.cadd/v1.7/GRCh38", ...]
  mane_gtf: ~/hf-rare/ref/gencode.v47.MANE.gtf.gz
  genome_fasta: ~/hf-rare/ref/GCA_000001405.15_GRCh38_no_alt_analysis_set.fna
  hp_obo: ~/hf-rare/ref/hp.obo
  hpo_map: ~/hf-rare/ref/hpo_gene_to_phenotype.tsv
  gnomad_v41_exomes_dir: ~/hf-rare/ref
  clinvar_vcf: ~/hf-rare/ref/clinvar.vcf.gz
variants.qc.min_gq: 20
  qc.min_dp: 10
  qc.min_alt_frac: 0.25
  af.max_rare: 0.01
  af.max_ultrarare: 0.0001
  effect.lof_terms: ["stop_gained", "frameshift", "splice_donor", "splice_acceptor"]
  effect.missense_terms: ["missense_variant"]
  effect.inframe_terms: ["inframe_deletion","inframe_insertion"]
  effect.exploit_cadd_if_available: true
pairing.compound_het_rule: "same_gene, >=2 founders, no hom-alt, phase-unknown-ok"
  same_gene_only: true
  allow_unknown_phase: true
phenotype.hpo_terms: ["HP:0002859", "HP:0000121", …] # eight terms
scoring.founder_weight: 0.45
  pathogenicity_weight: 0.60
  rare_weight: 0.30
  hpo_weight: 2.0
output.write_dir: ~/hf-rare/pipeline/output
  track1_csv_columns: ["proband_id","chrom_1",...]
```

---

6. Output contract

6.1 Ranked TSV (`annot_full.tsv`)

- **Location:** `output/annot_full.tsv`, as written by every documented invocation (which passes `--out output/annot_full.tsv`); `annot_ns.tsv` under `output.write_dir` is only the argparse default used when `--out` is omitted (`prioritize.py` line 85).
- **Columns (16), in file order (`prioritize.py` lines 197-198):**

#	Column	Notes
1	`rank`	1-based
2	`type`	`compound_het` or `hom_alt`
3	`gene`	
4-11	`chrom_1`, `pos_1`, `ref_1`, `alt_1`, `chrom_2`, `pos_2`, `ref_2`, `alt_2`	first and second allele coordinates, bare contig names
12	`epcr`	four-decimal float, clamped to (0, 1]
13-14	`consequence_1`, `consequence_2`	

#	Column	Notes
15-16	`cadd_1`, `cadd_2`	may be empty

- **Generation:** performed by `prioritize.py::main()` (writes a header line then rows).
- **Observed size:** 6,962 rows (see *REPORT.md* line 8).

6.2 Challenge-compliant CSV (`submission_track1.csv`)

- **Schema (12 columns):** `proband_id,chrom_1,pos_1,ref_1,alt_1,chrom_2,pos_2,ref_2,alt_2,epcr,finding_type,notes`
- **Enforcement:** `make_submission_csv.py` reads the TSV, adds the fixed `PROBAND01` identifier, prefixes chromosomes with `chr`, clamps EPCR, formats `epcr` to four decimals, determines `finding_type` (`primary` for compound-het, `secondary` for homozygous singles), and builds a compact `notes` string of the form `gene=...;consequence1[/consequence2]`.
- **Delivered file:** `submission_track1.csv` in the repository matches this contract (header line matches exactly, see *submission_track1.csv* line 1).

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7. Reproduction recipe

From a fresh checkout:

```
# 1. Create a virtual environment (as documented in README.md)
python3 -m venv .venv && source .venv/bin/activate
pip install pysam cyvcf2 pyyaml

# 2. Run the pipeline (full run, reproducing the reported numbers)
python3 pipeline/scripts/prioritize.py \
    --conf pipeline/conf/track1_config.yml \
    --with-af --with-hpo \
    --out output/annot_full.tsv

# 3. Convert to the challenge CSV (top 10 rows as submitted)
python3 pipeline/scripts/make_submission_csv.py \
    --in output/annot_full.tsv \
    --out submission_track1.csv
```

Expected run characteristics (as recorded in *REPORT.md*): - Wall-clock time: **1 h 31 min** on a **48-core** workstation (CPU-only). - QC passes **4,592,493** out of **5,012,204** input records. - After annotation **12,952** coding/splice candidates remain. - Pairing yields **2,211** compound-het pairs and **4,751** homozygous candidates, giving a **6,962**-row ranked table. - The top-ranked EPCR is **0.5823** (BUB1B pair).

The pipeline is deterministic: no random numbers, no date-dependent file names, and all sorting uses Python's stable `list.sort`.

This documentation itself is rebuilt by `docs/build_pdf.sh` (pandoc with the XeLaTeX engine and the DejaVu Serif main font - pdflatex cannot encode non-ASCII glyphs):

```
./docs/build_pdf.sh     # -> TRACK1_TECHNICAL_DOCUMENTATION.pdf
```

---

8. References

Tools & libraries

- **cyvcf2** - fast VCF parsing. Documentation: https://brentp.github.io/cyvcf2/
- **pysam** - Python bindings for HTSlib. Documentation: https://pysam.readthedocs.io/en/latest/
- **PyYAML** - YAML parsing. Documentation: https://pyyaml.org/wiki/PyYAMLDocumentation
- **matplotlib** - figure generation. Documentation: https://matplotlib.org/stable/
- **pandoc** - Markdown to PDF conversion. Documentation: https://pandoc.org/MANUAL.html

Primary literature (cited in the pipeline or report)

- Li, H. *et al.* **The Sequence Alignment/Map format and SAMtools**. *Bioinformatics* **25**, 2078-2079 (2009). DOI: https://doi.org/10.1093/bioinformatics/btp352

Pipeline overview: stages top to bottom; dashed lane = in-repo checks guarding the annotation and output steps.

Figure 1: Pipeline overview: stages top to bottom; dashed lane = in-repo checks guarding the annotation and output steps.

- Li, H. **Tabix: fast retrieval of sequence features from generic TAB-delimited files**. *Bioinformatics* **27**, 718-719 (2011). DOI: https://doi.org/10.1093/bioinformatics/btq671
- Chen, S. *et al.* **A genomic mutational constraint map using variation in 76,156 human genomes**. *Nature* **614**, 529-536 (2024). DOI: https://doi.org/10.1038/s41586-023-06045-0
- Kohler, S. *et al.* **The Human Phenotype Ontology in 2024**. *Nucleic Acids Res* **52**, D1332-D1343 (2024). DOI: https://doi.org/10.1093/nar/gkad1005
- Smedley, D. *et al.* **Exomiser: next-generation diagnostics and disease-gene discovery**. *Nat. Protoc.* **10**, 1524-1535 (2015). DOI: https://doi.org/10.1038/nprot.2015.124
- Landrum, M. J. *et al.* **ClinVar: improving access to variant interpretations and supporting evidence**. *Nucleic Acids Res* **46**, D1062-D1067 (2018). DOI: https://doi.org/10.1093/nar/gkx1153

Reference data sources

- **gnomAD v4.1** - https://gnomad.broadinstitute.org/ (variant frequency data, used via `vcf.py`).
- **GENCODE v47 MANE GTF** - https://www.gencodegenes.org/
- **MANE (Matched Annotation from NCBI & EMBL-EBI)** - https://www.ncbi.nlm.nih.gov/refseq/MANE/
- **GRCh38 no-alt analysis set FASTA** - NCBI Assembly: https://www.ncbi.nlm.nih.gov/assembly/GCA_000001405.15/
- **CADD v1.7** - https://cadd.gs.washington.edu/
- **ClinVar** - https://www.ncbi.nlm.nih.gov/clinvar/
- **Human Phenotype Ontology (HPO)** - https://hpo.jax.org/ (ontology file `hp.obo`).

Link audit note (2026-09-28): all URLs above were checked with `curl -sL`. The six `doi.org` links resolve to the correct articles (verified against the Crossref API at https://api.crossref.org/works/DOI); four of them (btp352, btq671, gkad1005, gkx1153) return HTTP 403 to automated requests because Oxford Academic blocks non-browser clients - they open normally in a browser. All other links returned HTTP 200.

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9. Appendix - figure 1

Figure 1: Flowchart of the Track 1 variant-prioritization pipeline. Blue boxes denote pipeline stages, orange boxes local reference inputs, and the green box the configuration file. Solid arrows indicate data flow; dashed arrows show configuration/reference feeding.

All numbers and code references have been cross-checked against the repository artifacts (grep-verified). The documentation is self-contained and reproducible.